

Individual Addendum: Aniya Facen **Name:** Aniya Facen **Team:** ILOVECODING **Date:** May 1, 2026 --- ## 1. Personal Contribution My contributions to the ILOVECODING capstone project focused on data quality assurance, exploratory visualization, statistical modeling validation, and business interpretation across all four milestones. - **Milestone 1 (Data Cleaning & Validation):** Conducted comprehensive missing value analysis across 34,121 observations; implemented outlier detection protocol using IQR method for leverage and returns; created data quality report documenting 0.4% loss rate; verified REIT entity identifiers against master database; resolved 23 duplicate REIT-month combinations. **14 hours** - **Milestone 2 (Exploratory Visualization & Diagnostics):** Created sector-level return distribution plots and time-series trend charts; analyzed return volatility by REIT size and sector; conducted normality tests (Shapiro-Wilk) revealing right-tail fat distribution; drafted findings report with interpretation of preliminary patterns. **13 hours** - **Milestone 3 (Model Validation & Robustness Testing):** Conducted outlier sensitivity analysis on full sample vs. outlier-trimmed sample; ran Breusch-Pagan test for heteroskedasticity; tested alternative lag specifications (Lag 1 only, Lag 1-2 only, etc.); verified DiD parallel-trends assumption via visual inspection of pre-2015 return paths. **14 hours** - **Milestone 4 (Business Recommendations & Risk Assessment):** Authored Conclusions & Recommendations section with sector allocation guidance; drafted comprehensive Risk Assessment & Limitations section detailing model assumptions, omitted variables, and external validity concerns; synthesized robustness findings into investment caveats. **14 hours**

Total Capstone Hours: ~55 hours --- ## 2. One Defended Methodological Decision **Decision:** Use Breusch-Pagan heteroskedasticity test rather than Newey-West standard errors for addressing volatility clustering. **Reasoning:** In Milestone 3, we faced a choice: address time-varying volatility (heteroskedasticity) in monthly returns either by (1) estimating Newey-West standard errors that account for serial correlation and heteroskedasticity, or (2) diagnosing heteroskedasticity first via formal testing, then applying entity-level clustering. Our rationale for entity-level clustering over Newey-West: - **Heteroskedasticity is not the binding constraint.** The Breusch-Pagan test ($\chi^2 = 14.2$, $p=0.003$) confirms heteroskedasticity exists, but the economic source is within-REIT variation (some REITs are consistently more volatile than others), not time-variation in aggregate volatility. This is an entity effect, not a time effect. - **Entity clustering directly targets within-REIT correlation.** Newey-West assumes time-varying heteroskedasticity and serially correlated errors—appropriate for aggregate shocks but less so for firm-level panel data where cross-sectional differences dominate. Our entity fixed effects already control for constant within-REIT volatility differences. - **Evidence from M2:** Sector-level volatility analysis revealed Industrial REITs have $\sigma \approx 4.1\%$ monthly while Office REITs have $\sigma \approx 6.2\%$ —systematic differences across entity types, not time-varying patterns. This motivated entity clustering over heteroskedasticity-

consistent (HC) formulas. This choice is principled and conservative: entity clustering produces wider confidence intervals than homoskedasticity-assuming OLS, so our beta result ($t=4.46$, $p<0.001$) is robust even under conservative adjustments. --- ## 3. One Key Limitation

***Limitation:** Beta is estimated using 12-month rolling correlation, introducing potential look-ahead bias and time-varying measurement error.

****Why It Matters:**** We compute beta for the period $[t, t+11]$ and assign it to month t . This creates a mechanical relationship: a REIT's beta in month t partially includes forward-looking information (returns in months $t+1$ through $t+11$). If past returns predict future returns (momentum) or if beta varies across market regimes, our beta measure becomes time-varying and noisy. Additionally, REITs with less than 12 months of history receive no beta estimate, creating missing data and potential selection bias (newer REITs excluded).

****Impact on Findings:****

- ****Beta coefficient may be attenuated.**** Measurement error in beta (from rolling-window noise and look-ahead component) biases regression coefficients toward zero. Our 61 bps estimate is a lower bound on the true beta-return relationship.
- ****Inference is less affected.**** t -statistics are robust to classical measurement error as long as error is mean-zero. Our $t=4.46$ remains statistically significant.
- ****Temporal dynamics are obscured.**** True beta may change over time (financial crisis $\rightarrow$ higher market sensitivity), but rolling-window averaging masks these regime shifts.

****How to Address:**** Alternative approaches—(1) using Fama-French factor models to extract systematic risk, (2) estimating dynamic beta with rolling windows of 24 or 36 months for more stability, or (3) explicitly modeling regime-switching beta—would improve measurement. This limitation explains why our recommendation emphasizes beta tilting (which is robust) over precise quantitative return predictions (which require more accurate beta measurement). --- ## 4. AI Audit Notes

****Task 1:** Breusch-Pagan heteroskedasticity test implementation

****Prompt:**** "Write Python code to perform Breusch-Pagan test for heteroskedasticity on panel regression residuals, including test statistic, p-value, and interpretation." -

****Output:**** AI provided code using `statsmodels' het_breuschpagan()` function with correct syntax.

****Verification:**** Ran against actual M3 regression residuals; compared test output to manual calculation (regress squared residuals on explanatory variables). Result matched: $\chi^2 = 14.2$ ($p=0.003$). Confirmed interpretation (reject homoskedasticity null).

****Changes:**** Added comments clarifying that heteroskedasticity detected $\neq$ need for HC standard errors (entity clustering sufficient given our clustering choice).

****Task 2:** Robustness testing workflow design

****Prompt:**** "Outline a systematic robustness testing plan for panel regression, including subsample analysis, outlier sensitivity, and alternative specifications." -

****Output:**** AI provided comprehensive checklist: (a) drop outliers, (b) time-period subsamples, (c) lag specifications, (d) sector subsamples.

****Verification:**** Implemented each step; compared results manually. Found leverage effect sign-reversal under outlier removal, which AI did not predict. This discovery informed our caveats about fragile leverage finding.

****Changes:**** Expanded outlier detection procedure beyond AI suggestion to

include mahalanobis distance (multivariate outliers) in addition to univariate IQR.

****Task 3: Risk assessment and limitations writing**** - ****Prompt:**** "Write a comprehensive section explaining model limitations, omitted variables, and external validity concerns for a fixed effects panel regression of REIT returns." - ****Output:**** AI generated boilerplate text covering standard limitations (omitted variables, endogeneity, sample selection). - ****Verification:**** Reviewed for accuracy; AI template was generic but accurate. Tailored to our specific context (REIT leverage, beta measurement, sample period 2000–2024). - ****Changes:**** Added specific examples (off-balance-sheet leverage, 2008 crisis structural break, survivorship bias in REIT data) rather than generic statements. ****Overall Assessment:**** AI provided helpful scaffolds for technical tasks and writing, but all robustness results and substantive interpretation were human-reviewed and verified against code outputs. No AI-generated numerical results incorporated without manual spot-checking. --- ****Status:**** ✓ COMPLETE ****Submission Date:**** May 1, 2026 by 11:59 PM EST